

# EcoLegis: Paperless Legislative Document Management System

**Proposed Title:** EcoLegis: A Digital Session Binder & Document Routing System **Target Client/LGU:** Sangguniang Panlungsod, City of Malolos

**Domain:** Municipal E-Governance & Resource Optimization **Prepared For:** Thesis / Capstone Advisory Committee

## Executive Summary & Concept Overview

**EcoLegis** is an enterprise municipal document management solution designed specifically for the weekly legislative sessions of the Sangguniang Panlungsod of Malolos City. The project replaces the high-volume paper printing of weekly session agenda packages, ordinance drafts, committee reports, and annexes with a secure, digital pre-session routing workflow and interactive digital session binders. By focusing purely on document logistics, paper waste reduction, and version comparison—without touching complex parliamentary voting logic—this project delivers measurable environmental and financial savings for the City Government of Malolos while remaining technical and feasible for a 6-month capstone timeline.

15,000+

PAGES SAVED / QTR

~85%

PRINTING COST CUT

100%

DIGITAL AUDIT TRAIL

< 3 Sec

OCR ARCHIVE SEARCH

## 1. STATEMENT OF THE PROBLEM

During weekly legislative sessions, the Office of the Vice Mayor and the Secretary to the Sanggunian print thick paper packets (ranging from 30 to over 100 pages each) for every City Councilor, Secretariat staff member, legal counsel, and department head.

- **Severe Paper & Monetary Waste:** Thousands of sheets of paper and costly toner cartridges are consumed weekly, only to be discarded when minor amendments or revisions are made to draft ordinances.
- **Manual & Delayed Distribution:** Department heads submit proposed resolutions physically or via email, requiring manual compilation and manual distribution prior to session deadlines.
- **Inefficient Historical Searching:** Looking up past enacted ordinances or session minutes from previous years requires manual inspection of physical file folders in the City Hall archives.

## 2. PROJECT OBJECTIVES

**General Objective:** To design and develop a web-based legislative document distribution and archival system that minimizes paper waste and optimizes document routing for Malolos City Hall weekly council sessions.

**Specific Objectives:**

- Develop a **Pre-Session Submission Module** allowing city departments to digitally submit proposed resolutions with automated deadline enforcement.
- Build an automated **Digital Session Binder Compiler** that generates structured, indexed digital session packages for Councilors.

- Implement **In-App Private Annotation Tools** allowing Councilors to highlight and take notes on session documents on-screen without altering master files.
- Integrate a **Redline Document Version Comparison Tool** to highlight text edits between ordinance drafts (v1.0 vs v1.1) to avoid re-printing entire packages.
- Establish an **OCR-Indexed Legislative Archive** for fast keyword searching of all historical Malolos City ordinances and resolutions.
- Provide an **Environmental & Financial Analytics Dashboard** tracking real-time metrics on paper reams saved, cost reductions (PHP), and carbon footprint offset.

### 3. CORE SYSTEM MODULES & FEATURES

| Module                            | Functional Capabilities                                                                                                   | User Roles Involved          |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------|
| 1. Document Routing & Submissions | Digital proposal uploads, automated submission cutoff timers, Secretariat review queue, status notifications.             | Dept Heads   Secretariat     |
| 2. Interactive Session Binder     | 1-click weekly agenda compilation, PDF viewer, digital sticky notes, line highlighting, encrypted offline local caching.  | Councilors   Vice Mayor      |
| 3. Versioning & Redline Diff      | Side-by-side version comparison, automated visual highlighting of added/deleted text between ordinance drafts.            | Secretariat   Councilors     |
| 4. Smart Ordinance Archival       | OCR text indexing, multi-filter keyword searching (by year, committee, author), public-facing read-only ordinance portal. | Public   Secretariat   Staff |
| 5. Eco-Analytics Dashboard        | Real-time computation of paper sheets saved, financial savings in PHP, trees saved metric, session-by-session trends.     | System Admin   LGU Officials |

### 4. SCOPE & DELIMITATIONS

To ensure project feasibility within the allocated academic timeline and maintain zero legal friction, the boundaries are strictly established as follows:

| In Scope (System Boundary)                                     | Out of Scope (Delimitations)                                                  |
|----------------------------------------------------------------|-------------------------------------------------------------------------------|
| <b>IN</b> Pre-session PDF/DOCX proposal submission & routing   | <b>OUT</b> Automated Parliamentary Procedure / Robert's Rules logic           |
| <b>IN</b> Interactive PDF viewing & personal in-app annotation | <b>OUT</b> Electronic or In-App Voting / Roll-call calculation                |
| <b>IN</b> Version history tracking & redline text comparisons  | <b>OUT</b> Legally binding PKI digital signing (physical signatures retained) |

| IN SCOPE (SYSTEM BOUNDARY)                                     | OUT OF SCOPE (DELIMITATIONS)                                    |
|----------------------------------------------------------------|-----------------------------------------------------------------|
| <b>IN</b> Public OCR search portal for enacted city ordinances | <b>OUT</b> Live video streaming or audio conference integration |

**Adviser Elevator Pitch:**

*"Ma'am, the Sangguniang Panlungsod prints over 15,000 pages of paper per quarter just for weekly session packets and revisions. Our system digitizes the entire pre-session document routing, digital reading, and archival workflow. This eliminates up to 85% of paper waste while providing Malolos City with an intelligent OCR archive and direct cost-saving analytics."*